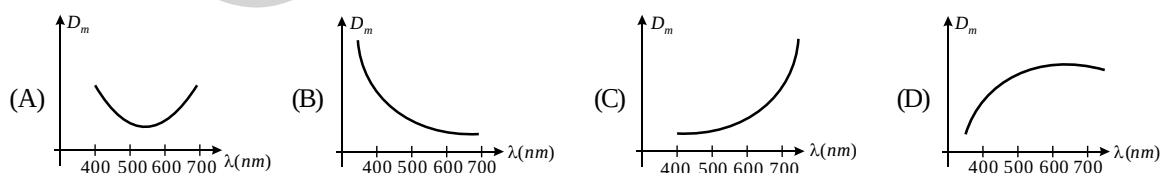
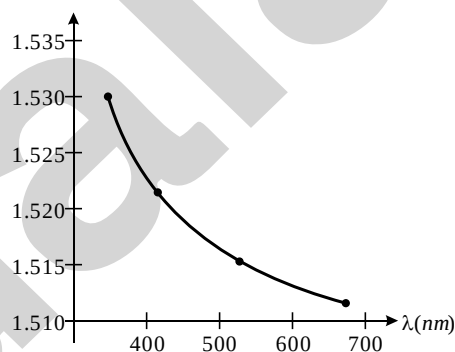


JEE Main Level Practice Test-16

for JEE & NEET Aspirants

Topic : RAY OPTICS
Time: 75Min Marking +4 -1
Section - A : MCQs with Single Option Correct

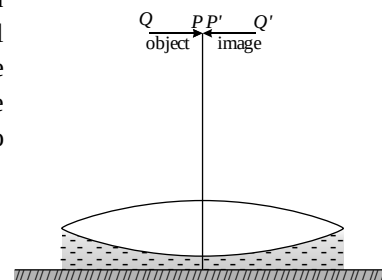
- The eye can be regarded as a single refracting surface. The radius of curvature of this surface is equal to that of cornea (7.5 mm). This surface separates two media of refractive indices 1 and 1.5. Calculate the distance from the refracting surface at which a parallel beam of light will come to focus :
 (A) 45mm (B) 22.5mm (C) 20mm (D) 15mm
- With a simple microscope, if the final image is located at infinity, then its magnifying power is :
 (A) $\frac{D}{f}$ (B) $1 + \frac{D}{f}$ (C) $\frac{f}{D}$ (D) $D \times f$
- If diameter of objective lens of a telescope is 4 cm and wavelength of light is 600 nm. The resolving power of telescope is :
 (A) 18.3×10^8 (B) 5.46×10^4 (C) 5.46×10^6 (D) 18.3×10^9
- A convergent doublet of separated lenses, corrected for spherical aberration, has resultant focal length of 10 cm. The separation between the two lenses is 2 cm. The focal lengths of the component lenses :
 (A) 18 cm, 20 cm (B) 10 cm, 12 cm (C) 12 cm, 14 cm (D) 16 cm, 18 cm
- The variation of refractive index of a crown glass thin prism with wavelength of the incident light is shown. Which of the following graphs is the correct one, if D_m is the angle of minimum deviation?



- A real object approaches a fixed diverging lens with a constant velocity from infinity along the principal axis. The relative velocity between object and its image will be :
 (A) increasing (B) decreasing
 (C) first increases then decreases (D) first decreases and then increases

7. The figure shows an equiconvex lens (of refractive index 1.50) in contact with a liquid layer on top of a plane mirror. A small needle PQ with its tip on the principal axis is moved along the axis until its inverted image is found at the position of the needle. The distance of the needle from the lens is measured to be 45.0 cm. The liquid is removed and the experiment is repeated. The new distance is measured to be 30.0 cm. The refractive index of the liquid is :

(A) $5/4$
(B) $9/8$
(C) $4/3$
(D) $6/5$



8. A convexo-concave diverging lens is made of glass of refractive index 1.5 and focal length 24 cm. Radius of curvature for one surface is double that of the other. Then radii of curvature for the two surfaces are (in cm) :

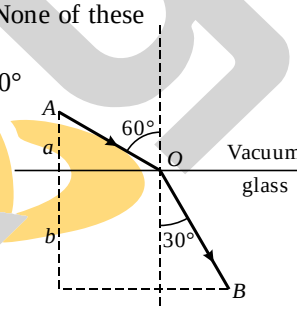
(A) 6, 12 (B) 12, 24 (C) 3, 6 (D) 18, 36

9. When a lens of power P (in air) made of material of refractive index μ is immersed in liquid of refractive index μ_0 . Then the power of lens is :

(A) $\frac{\mu-1}{\mu-\mu_0} P$ (B) $\frac{\mu-\mu_0}{\mu-1} P$ (C) $\frac{\mu-\mu_0}{\mu-1} \cdot \frac{P}{\mu_0}$ (D) None of these

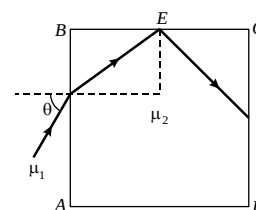
10. A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is :

(A) $2a + 2b$ (B) $2a + \frac{2b}{3}$
(C) $\frac{2\sqrt{3}}{a} + 2b$ (D) $2a + \frac{2b}{\sqrt{3}}$



11. A transparent cube of side d , made of a material of refractive index μ_2 , is immersed in a liquid of refractive index μ_1 ($\mu_1 < \mu_2$). A ray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC . The θ must satisfy :

(A) $\theta < \sin^{-1} \frac{\mu_1}{\mu_2}$ (B) $\theta < \sin^{-1} \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$
(C) $\theta > \sin^{-1} \frac{\mu_1}{\mu_2}$ (D) $\theta > \sin^{-1} \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$

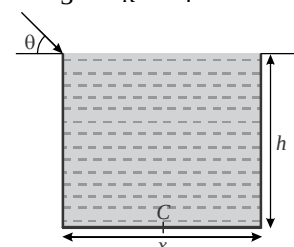


12. A bird in air looks at a fish directly below it inside a transparent liquid in a tank. If the distance of the fish as estimated by the bird is h_1 and that bird as estimated by the fish is h_2 , then the refractive index of liquid is :

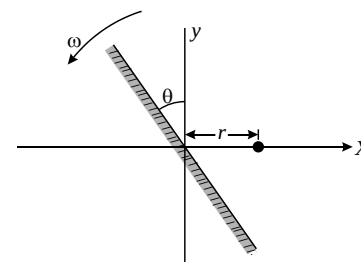
(A) $\frac{h_2}{h_1}$ (B) $\frac{h_1}{h_2}$ (C) $\frac{h_1 + h_2}{h_1 - h_2}$ (D) $\frac{h_1 - h_2}{h_1 + h_2}$

13. A student sees the top edge and the bottom center C of a pool simultaneously from an angle θ above the horizontal as shown in the figure. The refractive index of water which fills up to the top edge of the pool is $\frac{4}{3}$. If $\frac{h}{x} = \frac{7}{4}$ then $\cos \theta$ is :

(A) $\frac{2}{7}$ (B) $\frac{8}{3\sqrt{45}}$
(C) $\frac{8}{3\sqrt{53}}$ (D) $\frac{8}{21}$



14. The plane mirror shown is rotated with constant angular velocity ω about origin in xy plane as shown. Object is placed at a distance r from origin on x -axis. Find velocity of image with respect to origin of coordinate system at an instant when mirror makes an angle θ with +ve y -axis as shown :



- (A) $2r\omega \sin 2\theta i - 2r\omega \cos 2\theta j$
 (B) $2r\omega \cos 2\theta i + 2r\omega \sin 2\theta j$
 (C) $r\omega \sin \theta i + r\omega \cos \theta j$
 (D) None of these

15. The minimum length of a plane mirror to see the entire full-length image of an object is half of the object's height. Suppose δ is the distance between eye and top of the head of a person of height h . The person will be able to see his entire full-length image with a mirror of height $h/2$ fixed on the wall :

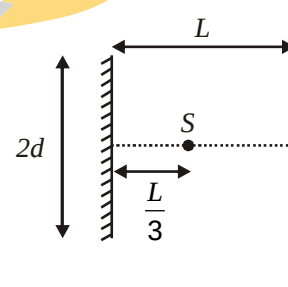
- (A) when the bottom edge of mirror is kept $h/2$ above the floor.
 (B) when the bottom edge of mirror is kept $(h + \delta)/2$ above the floor.
 (C) when the bottom edge of mirror is kept $(h - \delta)/2$ above the floor.
 (D) when the centre of the mirror is at the same height as centre of the person.

16. A certain prism is found to produce a minimum deviation of 38° . It produces a deviation of 44° when the angle of incidence is either 42° or 62° . What is the angle of incidence when it is undergoing minimum deviation?

- (A) 45° (B) 49° (C) 40° (D) 55°

17. A point source of light S is placed at a distance $L/3$ in front of the center of plane mirror of width $2d$ which is hanging vertically on a wall. A man walk's in front of mirror, at a distance L as shown below. The distance over which the man can see the image of the light source in the mirror is :

- (A) $8d$
 (B) $4d$
 (C) $\frac{5d}{8}$
 (D) $\frac{3d}{2}$

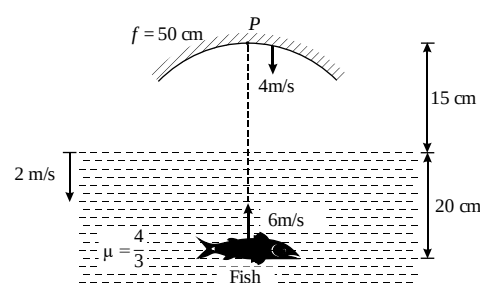


18. A light ray along $2\hat{i} + 3\hat{j} + 4\hat{k}$ incident and reflected from a plane mirror placed along YZ plane. What is the unit vector along the reflected ray?

- (A) $\frac{-2\hat{i} + 3\hat{j} + 4\hat{k}}{\sqrt{29}}$ (B) $\frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}}$ (C) $\frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{\sqrt{29}}$ (D) $\frac{-2\hat{i} - 3\hat{j} - 4\hat{k}}{\sqrt{29}}$

19. In the figure shown a concave mirror of focal length 50 cm is made to move downward with constant velocity of 4 m/s. The level of water having $\mu = 4/3$ is also falling downward at speed of 2 m/s. Inside the water the fish is moving upward with speed of 6 m/sec and is lying along principal axis of concave mirror. Find out the speed of image of fish which is formed after reflection through concave mirror at the instant shown :

- (A) 36 m/s downward
 (B) 54 m/s downward
 (C) 18 m/s downward
 (D) None of these

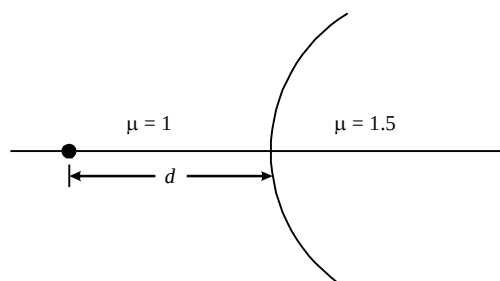


20. Two mirrors are inclined at an angle 60° . Find out the angle of incidence of a light ray which incidents on one mirror and becomes parallel to first mirror after 2 reflection :

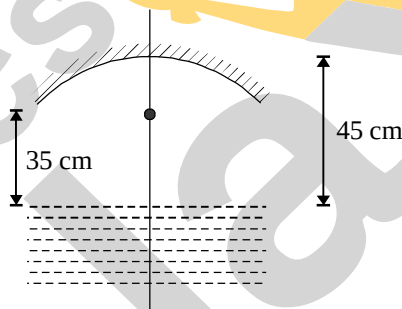
- (A) 20° (B) 30° (C) 40° (D) 50°

Section- B: INTEGER Answer Type Questions

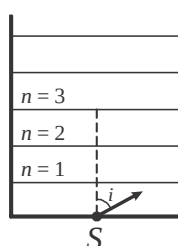
21. A Galilean telescope measures 10 cm from the objective to the eyepiece. The focal length of the objective is 20 cm. Calculate the magnifying power of the telescope.
22. A point object is placed in air at a distance d from the pole of a refracting convex surface on its principal axis & at $t = 0$ it starts moving with speed c perpendicular to its principle axis as shown figure. The radius of the convex surface is d . Calculate the ratio of speed of image to speed of object at $t = 0$.



23. A concave mirror of focal length 15 cm is placed in air at a height of 45 cm above the water surface as shown. The axis of the mirror is vertical and the reflecting surface faces the water surface. A point object is placed on the axis of concave mirror at a distance 35 cm above the water surface. Find the position of image formed after two successive processes- first reflection from the concave mirror and then first refraction from the water (refractive Index of water $4/3$). Calculate the distance of image from water surface (in m).

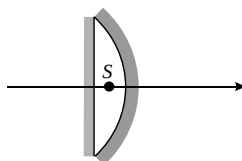


24. A point source S is placed at the bottom of different layers as shown in figure. The refractive index of bottom most layer is μ_0 . The refractive index of n^{th} layer from bottom as shown in figure is given as $\mu(n) = \mu_0 - \frac{\mu_0}{5n-18}$ where $n = 1, 2, 3, \dots$. A ray of light with $i = 30^\circ$ starts from the source S . The total internal refraction takes place at the upper surface of n^{th} layer. Calculate value of n .

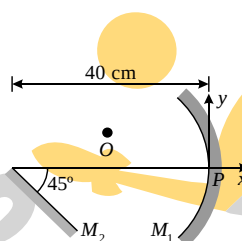


25. The focal length of a concave mirror is 20 cm. When an object is moved from a distance of 25 cm in front of it to 50 cm, the magnification of its image changes from m_{25} to m_{50} . Calculate the ratio $\frac{m_{25}}{m_{50}}$.

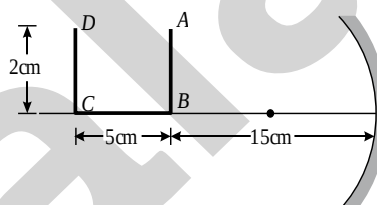
26. In the figure shown a thick plano convex lens is silvered at both the surfaces. A point source of light 'S' is inside the lens. The radius of curvature of the curved surface is $\frac{8}{3}$ cm and the distance of 'S' from the curved surface is 2 cm. If the images formed due to direct reflections from the plane and curved surface coincide then find the thickness (in cm) of the lens at the middle.



27. A lens with a focal length of 16 cm produces a sharp image of a real object in two positions of lens which are 60 cm apart. Find the distance from the object to the screen (in m).
28. As shown in the figure, an object O is at the position $(-10 \text{ cm}, 2 \text{ cm})$ with respect to the origin P. The concave mirror M_1 has radius of curvature 30 cm. A plane mirror M_2 is kept at a distance 40 cm in front of the concave mirror. Considering first reflection on the concave mirror M_1 and second on the plane mirror M_2 . The Y coordinates of the second image w.r.t. the origin P is $-10n$ cm. Find n .



29. A U-shaped wire is placed before a concave mirror having radius of curvature 20 cm as shows in figure the total length of the image is $2(x)$. Find the value of x .



30. For a glass prism ($\mu = \sqrt{2}$), the angle of minimum deviation is equal to the angle of the prism. Calculate the angle (in degree) of the prism.

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JEE Main Level Practice Test-16
for
JEE & NEET Aspirants**Topic : RAY OPTICS****ANSWER KEY****Section - A : MCQs with Single Option Correct**

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|---------|---------|---------|---------|
| 1. (B) | 2. (A) | 3. (B) | 4. (A) |
| 5. (B) | 6. (B) | 7. (C) | 8. (A) |
| 9. (C) | 10. (A) | 11. (B) | 12. (A) |
| 13. (C) | 14. (D) | 15. (C) | 16. (B) |
| 17. (A) | 18. (A) | 19. (B) | 20. (B) |

Section- B: INTEGER Answer Type Questions

- | | | | |
|---------|----------|---------|---------|
| 21. [2] | 22. [2] | 23. [1] | 24. [4] |
| 25. [6] | 26. [3] | 27. [1] | 28. [7] |
| 29. [8] | 30. [90] | | |

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